

CS 105: Department Introductory Course on Discrete Structures

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Lecture 13 – Counting and Combinatorics

Summary till now

Course Outline

1. Proofs and reasoning

- ▶ Propositions, predicates
- ▶ Proofs and proof techniques: contradiction, contrapositive, (strong) induction, well-ordering principle, diagonalization.

2. Basic discrete structures

- ▶ (finite and infinite) sets
- ▶ Functions: injections, surjections, bijections
- ▶ Relations: equivalence relations, partial orders, lattices
- ▶ Some applications

3. Counting and combinatorics

4. Introduction to graph theory

New Chapter: Counting and Combinatorics

Topics to be covered

- ▶ Last lecture before midsem: Basics of counting
 - ▶ Product principle
 - ▶ Sum principle
 - ▶ Bijection principle
 - ▶ Double counting
- ▶ Today: Subsets, partitions, Permutations and combinations
 1. Binomial coefficients and Binomial theorem
 2. Pascal's triangle
 3. Permutations and combinations with repetitions
- ▶ Recurrence relations and generating functions
- ▶ Pigeonhole Principle and its extensions

An interesting example with double counting

Handshake Lemma

At a meeting with n people, the number of people who shake hands an odd number of times is even.

Proof in six steps:

1. Define relation R : iRj if i and j shook hands. Is it symmetric (trans/refl.)?
Draw its graph!
2. Let m_i be the number of times i shakes hands. i.e., m_i is the number of directed edges going out from i . So, **Total no. of directed edges** $= \sum_{i=1}^n m_i$.
3. Now, let X be the total number of handshakes. Clearly this is an integer.
Total no. of directed edges $= 2 \cdot X$.
4. i.e., $2 \cdot X = \sum_{i=1}^n m_i = \sum_{\{i | m_i \text{ is odd}\}} m_i + \sum_{\{i | m_i \text{ is even}\}} m_i$
5. $\sum_{\{i | m_i \text{ is even}\}} m_i$ is even, as sum of even numbers is even.
So $\sum_{\{i | m_i \text{ is odd}\}} m_i$ must also be even, but each of these m_i is odd, so $|\{i \mid m_i \text{ is odd}\}|$ has to be even.
6. Thus number of i such that m_i is odd is even! □

Moving on: Binomial theorem

Recall: $\sum_{k=0}^n \binom{n}{k} = 2^n$.

We generalize this...

Binomial Theorem

Let x, y be variables and $n \in \mathbb{Z}^{\geq 0}$. Then,

$$(x + y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j$$

$$(x + y)^1 = x + y$$

$$(x + y)^2 = (x + y)(x + y) = x^2 + 2xy + y^2$$

$$(x + y)^3 = (x + y)(x + y)^2 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x + y)^4 = (x + y)(x + y)^3 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

...

Moving on: Binomial theorem

Binomial Theorem

Let x, y be variables and $n \in \mathbb{Z}^{\geq 0}$. Then,

$$(x + y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j$$

(H.W-1) Prove this by induction.

Moving on: Binomial theorem

Binomial Theorem

Let x, y be variables and $n \in \mathbb{Z}^{\geq 0}$. Then,

$$(x + y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j$$

Proof (Combinatorial):

1. Consider any term $x^i y^j$, where $i + j = n$.
2. To get $x^i y^j$ term in

$$(x + y)(x + y) \cdots (x + y) \quad (n \text{ times})$$

we need to pick j y 's from n sums and remaining x 's.

3. Thus, the coefficient of this term = number of ways to get this term = number of ways to pick j y 's from n elts = $\binom{n}{j}$.

Moving on: Binomial theorem

Binomial Theorem

Let x, y be variables and $n \in \mathbb{Z}^{\geq 0}$. Then,

$$(x + y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j$$

Corollaries Ex: Using Binomial Thm show the following

1. $\binom{n}{j} = \binom{n}{n-j}$,
2. $\sum_{j=0}^n \binom{n}{j} 2^j = 3^n$.
3. No. of subsets of n -element set having even cardinality = No. of subsets of n -element set having odd cardinality (=?)

Pascal's Triangle

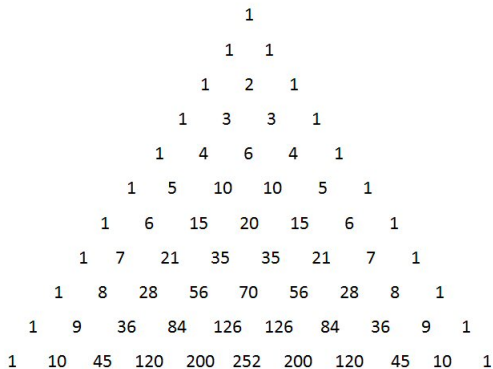
A recursive way to compute binomial coefficients

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

$$\begin{array}{c} \binom{0}{0} \\ \binom{1}{0} \quad \binom{1}{1} \\ \binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2} \\ \binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3} \\ \binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4} \\ \binom{5}{0} \quad \binom{5}{1} \quad \binom{5}{2} \quad \binom{5}{3} \quad \binom{5}{4} \quad \binom{5}{5} \end{array}$$

$$\begin{array}{ccccccc} & & & & 1 & & \\ & & & 1 & & 1 & \\ & & 1 & & 2 & & 1 \\ & 1 & & 3 & & 3 & & 1 \\ 1 & & 4 & & 6 & & 4 & & 1 \\ 1 & & 5 & & 10 & & 10 & & 5 & & 1 \end{array}$$

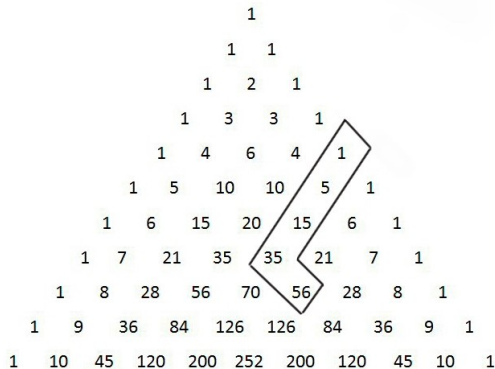
Fun with Pascal's triangle



Some simple observations. Recall: $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

1. Row i adds up to 2^i , Row $i + 1$ adds up to twice of row i .
2. Sequence of numbers, squares, cubes?

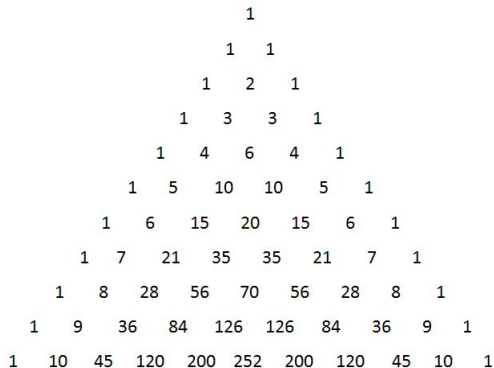
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Some simple observations. Recall: $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

1. Row i adds up to 2^i , Row $i + 1$ adds up to twice of row i .
2. Sequence of numbers, squares, cubes?
3. Hockey stick patterns: (H.W-2) $\binom{n+1}{m} = \binom{n}{m} + \binom{n-1}{m-1} \dots + \binom{n-m}{0}$

Fun with Pascal's triangle



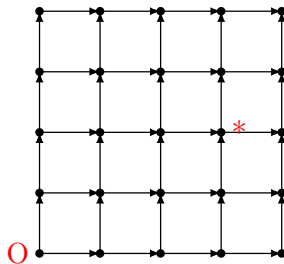
Some **not so** simple observations

- ▶ For some rows, all values in row (except first and last) are divisible by second!
- ▶ In fact, for all prime rows? why should p divide $\binom{p}{r}$, $r < p$?
- ▶ **Corollary:** $2^p - 2$ is a multiple of p , for any prime p .
- ▶ **Interesting Ex.:** Count no. of odd numbers in each row...

An application to path counting

Map problems

From the bottom left, how many shortest routes lead to a particular junction?



- ▶ Denote path sequences of $\{U, R\}$, e.g., $RRRUU$ reaches $*$.
- ▶ Our path has 5 segments of which there must be 3 R's.
- ▶ Bijection from set of paths to subsets of fixed size.
- ▶ No. of paths = ways of choosing 3 R's out of 5 elts = $\binom{5}{3}$.

Permutations and Combinations with repetitions

How many ways can you select k objects from a set of n elements?

- ▶ Depends on whether order is significant:
 - ▶ If order matters, then permutations
 - ▶ Else, combinations.
- ▶ What if repetitions are allowed?

	Order significant	Order not significant
Repetitions not allowed	$n(n-1)\dots(n-k+1)$	$\binom{n}{k}$
Repetitions are allowed	n^k	??

Combinations with repetitions

Theorem

The no. of ways k objects can be chosen from n , when repetition is allowed is $\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$.

1. Represent them as a list of $n + k - 1$ comprising of two types of symbols: bars (separators) and stars (objects).
 2. e.g., suppose we want to select 5 obj from a set of 4 with repetitions. Then, $** | * || **$ represents: 2 of first, 1 of second, none of third and 2 of fourth.
 3. There is a bijection between such lists and k -sets of n -obj with repetitions allowed (why?).
 4. Thus, question reduces to no. of ways to choose k stars or $n - 1$ bars from a set of $n - k + 1$ positions $= \binom{n+k-1}{k}$. □
- **Ex:** How many solutions does the equation $x_1 + x_2 + x_3 + x_4 = 17$ have such that $x_1, x_2, x_3, x_4 \in \mathbb{Z}^{\geq 0}$?

Estimating $n!$

How big is $n!$?

- ▶ It is clearly bigger than n and n^2 .
- ▶ Is it bigger than 2^n , n^n ?
- ▶ Easy to see: for all $n \geq 4$,

$$2^n \leq n! \leq n^n$$

- ▶ Can we do better?

Can we quantify how much more n^n is compared to $n!$?

Theorem (Stirling's Approximation)

$$e \left(\frac{n}{e}\right)^n \leq n! \leq ne \left(\frac{n}{e}\right)^n$$

where e is the base of natural logarithms, $\log(e) = e^{\log(e)} = 1$.